Week 4, Day 3: Feature Engineering, Cross Validation & Model Comparison

.

1. Engineered Feature List

Feature	Type	Rule	Justification	MI Score
age_bucket	Categorical (binned)	pd.cut(age, fixed bins)	Income rises then plateaus with age, non-monotonic	0.0728
hours_bucket	Categorical (binned)	pd.cut(hours-per-week, fixed bins)	Part time / standard / overtime likely pay differently	0.0449
has_capital_gain	Binary flag	capital-gain > 0	Capital gain acts as a threshold signal, not continuous	0.0293
has_capital_loss	Binary flag	capital-loss > 0	Any recorded loss may signal investment activity	0.0076
higher_education	Binary flag	education-num >= 13	Bachelor's or higher is a natural, interpretable cutoff	0.0479
log_capital_gain	Numeric transform	log1p(capital-gain)	Tames right skew for tree models specifically	0.0805
edu_hours_interaction	Interaction	education-num * hours-per-week	Captures compounding effect of education and hours	0.0835
net_capital	Numeric	capital-gain - capital-loss	Nets gains and losses into one investment signal	0.1180

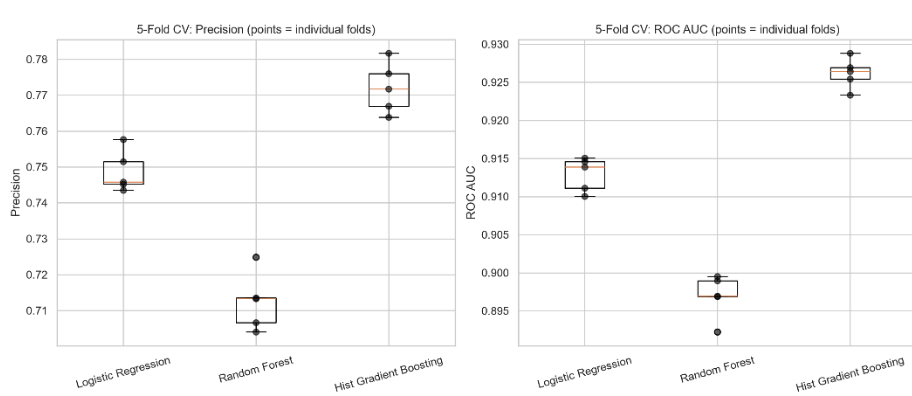
MI score = mutual_info_classif against the target (age_bucket and hours_bucket scores are summed across their one-hot dummy columns). net_capital carries the strongest univariate signal of any engineered feature, ahead even of the raw capital-gain/capital-loss columns individually, confirming that netting gains against losses captures more predictive information than either alone. has_capital_loss is the weakest, likely because so few rows have any recorded loss at all, limiting how much it can distinguish.

2. Cross Validated Model Comparison

A FunctionTransformer wrapping the feature engineering step feeds a three-branch ColumnTransformer (numeric, binary, categorical), refit inside every StratifiedKFold(k=5) split so no fold's validation rows influence another fold's training statistics. Three models were compared with identical preprocessing.

Model	Precision (mean ± std)	Recall (mean ± std)	F1 (mean ± std)	ROC AUC (mean ± std)	Fit time
Logistic Regression	**0.7488 ± 0.0052**	0.6149 ± 0.0086	0.6752 ± 0.0061	0.9129 ± 0.0020	1.8s
Random Forest	**0.7125 ± 0.0072**	0.6242 ± 0.0065	0.6654 ± 0.0063	0.8969 ± 0.0026	13.1s
Hist Gradient Boosting	**0.7720 ± 0.0063**	0.6545 ± 0.0092	0.7084 ± 0.0071	0.9262 ± 0.0018	4.0s

Hist Gradient Boosting is the strongest model, achieving the highest **precision (0.7720)**, **F1-score (0.7084)**, and **ROC AUC (0.9262)** while remaining substantially faster than Random Forest. Logistic Regression provides competitive performance with the benefit of interpretability, whereas Random Forest is both slower and weaker across all reported metrics. All three models exhibit low fold-to-fold variance, indicating stable cross-validation performance..



3. Statistical Test Results

Comparison	Mean Precision diff	Paired t-test	Wilcoxon
Hist GB vs. Logistic Regression	0.0233	t = 6.377, p = 0.0031	W = 0.000, p = 0.0625

Hist Gradient Boosting exceeds Logistic Regression by a mean **precision** difference of **0.0233**, representing a meaningful improvement under this project's primary metric. The paired t-test indicates this difference is statistically significant (p = 0.0031). Although the Wilcoxon test reports p = 0.0625, this is expected with only five paired folds; W = 0.000 shows Hist Gradient Boosting outperformed Logistic Regression on every fold. Taken together, the statistical tests indicate a genuine and practically meaningful improvement rather than random variation.

Feature importance and coefficients (engineered features only)

Feature	RF Importance	LogReg Coefficient
edu_hours_interaction	0.0960	-0.2030
net_capital	0.0537	2.4833
log_capital_gain	0.0413	-0.1542
higher_education	0.0291	0.0482
has_capital_gain	0.0117	-1.7902
has_capital_loss	0.0056	-2.2336
age_bucket (sum of dummies)	0.0459	range: -0.95 to 0.46
hours_bucket (sum of dummies)	0.0246	range: -0.86 to 0.33

edu_hours_interaction is Random Forest's single most important engineered feature (0.096), nearly double the next highest, consistent with trees being well suited to exploit a genuine multiplicative interaction that a single linear coefficient cannot represent directly. Logistic Regression, in contrast, assigns it a small negative coefficient, and shows a similar apparent contradiction for net_capital (strongly positive, 2.48) against has_capital_gain and has_capital_loss (both strongly negative). This is not evidence the features mean something different to each model, it reflects two compounding statistical effects rather than one. First, multicollinearity: net_capital, log_capital_gain, has_capital_gain, and has_capital_loss are all derived from the same underlying capital-gain/capital-loss columns, and edu_hours_interaction is a product of two features (education-num, hours-per-week) that remain in the model separately, so correlated predictors cause Logistic Regression to distribute credit for shared signal in ways that can flip individual signs while leaving overall predictions accurate. Second, Random Forest's default importance metric (mean decrease in impurity) is known to favor continuous, high-cardinality features over binary flags, which independently explains why has_capital_gain and has_capital_loss score low in the RF ranking even though they carry real, interpretable signal, this is a property of the importance metric itself, not evidence the flags are uninformative. Both models agree, despite these differences, that prime working-age brackets (36-55) and longer hours worked (41+) carry a positive, directionally sensible signal.

4. Feature Selection / Dimensionality Check

Configuration	Precision (mean ± std)	Recall (mean ± std)	F1 (mean ± std)	ROC AUC	Fit time
Full feature set	**0.7720 ± 0.0063**	0.6545 ± 0.0092	0.7084 ± 0.0071	0.9262	9.5s
SelectKBest (k=65)	**0.7705 ± 0.0027**	0.6432 ± 0.0030	0.7011 ± 0.0014	0.9251	19.9s

SelectKBest (mutual information) was evaluated against the full engineered feature set using **k = 65**. Precision decreases only slightly (0.7720 → 0.7705), while recall and F1-score also decline, and training time more than doubles because `mutual_info_classif` must be recomputed inside every cross-validation fold. Since feature selection neither improves the project's primary metric nor reduces computational cost, the full engineered feature set remains the preferred choice.

5. Recommendation for Day 4 Tuning

- **Recommended model:** Histogram Gradient Boosting, achieving the highest **precision (0.7720)**, strongest **ROC AUC (0.9262)**, and a consistent fold-by-fold advantage over Logistic Regression.
- **Recommended feature set:** The complete engineered feature set (5 raw numeric, 7 raw categorical, 8 engineered). Feature selection did not improve precision or training efficiency, so no reduction is recommended before hyperparameter tuning.